KENYA MEDICAL TRAINING COLLEGE
DEPARTMENT OF MEDICAL ENGINEERING

DIPLOMA IN MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION

ELECTRONICS

Time: 3 HOURS.

Instructions to the candidates

- This paper consists of **SEVEN** questions, answer any **FIVE** questions.
- All questions carry equal marks.
- You should have the following for this examination;
  - *i) Non programmable calculator.*
  - *ii) Drawing instruments.*
  - *iii) Answer booklet, which is provided.*
- Maximum marks for each part of a question are shown.

This paper consists of three (3) printed pages

© 2018, Kenya Medical Training College

Scanned with CamScanner

1. a) State three basic logic gates (3 marks)
   b) Determine the compliment of $F = A\bar{B}C + \bar{A}BD$ (4 marks)
      (i) Using de-Morgan's rules (4 marks)
      (ii) Using a Karnaugh map
   c) Use 1st complement and 2nd complement arithmetic to calculate (5 marks)
      (i) (27) (4 marks)
      (ii) (722)

2. a) State any three applications of a cathode Ray Oscilloscope (3 marks)
   b) A transistor amplifier uses transformer coupling for amplication. The impedance of the transistor amplifier is 10kΩ while the input impedance of the next stage is 3kΩ. Determine the transformer ratio for perfect impedance matching. (4 marks)
   c) (i) State three necessary requirements for Oscillations to occur in an Oscillator
      (ii) With aid of a labeled circuit diagram describe the construction and operation of a tuned collector oscillator (10 marks)

3. a) Use 74HC151 8-to-1 multiplexer to implement the logic function $F = AB + \bar{B}C + \bar{A}BC$ (10 marks)
   b) Write down the truth tables for the D and J.K flip flops. (10 marks)

4. a) State any two application of an Bistable mutivibrator (2 marks)
   b) With aid of a labeled diagram, explain the principle of operation of a bistable transistor multivibrator (12 marks)
   c) In an astable multivibrator $R_1=R_2=10K\Omega$, $C_1=C_2=0.01\,\mu F$. Calculate the frequency of oscillation (6 marks)

5. a) State any three methods used in fault diagnosis (3 marks)
   b) Briefly explain the action of an LED. List any four advantages and one disadvantage of the LED as a display device (10 marks)
   c) An LED is to be connected between the output of a TTL devices and +5V. Determine the value of the series current-limiting resistor required. The LED on voltage is 1.6V and the low output voltage of the TTL circuit is 0.2V. Find also the power dissipated in the series resistance. Assume the safe LED current to be 20mA (7 marks)

6. a) With aid of diagrams, explain the construction and operation of a unijunction transistor (10 marks)
   b) A unijunction transistor has 10V between the bases. If the intrinsic stand-off ration is 0.65, find the value of standoff voltage. What will be the peak point if the forward voltage drop in the pn junction is 0.7V? (5 marks)
   c) Explain using a diagram how power is controlled to a lead using SCR (silicon controlled rectifier) (5 marks)

Scanned with CamScanner